

Chapter 12

Database

A **database** is an organized collection of data that allows us to **store, manage, retrieve, and update information** easily.

Uses of Database

- Stores large amounts of data
- Makes data retrieval easy
- Allows data to be updated and deleted
- Provides data security
- Reduces duplicate data
- Allows multiple users to access data

Examples of Database Systems

- MySQL
- Oracle
- PostgreSQL
- Microsoft SQL Server
- MongoDB

2. MySQL

MySQL is an **open-source Relational Database Management System (RDBMS)** that uses **SQL (Structured Query Language)** to store and manage data.

It stores data mainly in the form of **tables**, which contain rows and columns.

Example

```
CREATE DATABASE college;
```

```
USE college;
```

```
CREATE TABLE student (
```

```
    id INT,
```

name VARCHAR(50),

);

Insert data:

```
INSERT INTO student VALUES (101, 'Arun', 85);
```

Advantages of MySQL

Open-source and free

Easy to learn

Fast and reliable

Supports large databases

Provides security

Supports multiple users

Works with languages such as Java, PHP and Python

3. Parts of MySQL

The main parts/components involved in MySQL are:

1. MySQL Server

The MySQL Server is the main component that manages databases, tables, queries, users, and data.

2. MySQL Client

The client is used to connect to the MySQL server and execute SQL commands.

Example:

```
SELECT * FROM student;
```

3. MySQL Database

A database is a collection of related tables.

Example:

College Database

|

└── Student

4. Tables

Tables store actual data in rows and columns.

Student

ID | Name | Mark

1 | Arun | 85

2 | Ravi | 78

5. SQL

SQL (Structured Query Language) is used to communicate with the database.

Examples:

CREATE

INSERT

SELECT

UPDATE

DELETE

MySQL Storage Engine

Storage engines determine how MySQL stores and retrieves data.

Common examples:

- **InnoDB**
- **MyISAM**

InnoDB is commonly used because it supports transactions and foreign keys.

4. MySQL Operators

Operators are symbols or keywords used to perform operations on data.

The major types are:

1. Arithmetic operators
2. Comparison operators

Assignment operators

- 3.
- 4.
- 5.
6. Special operators